

Beginner Guide: AI Image Generation APIs (OpenAI Example)

Audience: Complete beginner / fresher

Goal: Understand how AI image APIs work, create an account, get an API key, estimate costs, and build your own API.

1. What is an AI Image Generation API?

An AI Image Generation API is a service that creates a new image from a text prompt. You send a prompt (for example: 'A tiger walking in snow') to the API and it returns a generated image.

2. Why use an existing API?

- Fast to build
- No GPU required
- High quality
- Good for learning and prototypes

3. Create an OpenAI Account

1. Visit <https://platform.openai.com>
2. Sign up or sign in.
3. Open the Platform dashboard.

4. Add Billing

Open Settings → Billing.

Add a supported payment method.

Note: ChatGPT subscription and API billing are separate.

5. Create an API Key

Go to API Keys → Create new secret key.

Copy the key and store it safely.

Never expose it in frontend code or GitHub.

6. Install Python Library

```
pip install openai
```

7. Basic Workflow

User → Django Backend → OpenAI Image API → Generated Image → Save → Show to User

8. Approximate Costs

Typical planning estimates (1024x1024):

- Low quality: about ₹1 per image
- Medium quality: about ₹3.5–₹4 per image

- High quality: about ₹14–₹15 per image
Actual pricing may change over time.

9. Example Monthly Cost

100 medium images $\approx$ ₹400

1,000 medium images $\approx$ ₹4,000

10,000 medium images $\approx$ ₹40,000

10. Build Your Own API

Create endpoints such as:

POST /api/v1/image/generate

GET /api/v1/history

Your backend calls the AI provider and returns the result.

11. Later: Host Your Own Model

After your product grows, you can host open-source models (such as FLUX or Stable Diffusion) on your own GPU server to reduce recurring API costs.

12. Recommended Learning Roadmap

Week 1: Learn REST APIs and prompts.

Week 2: Integrate OpenAI Image API.

Week 3: Build Django API.

Week 4: Learn Docker, Celery, Redis, AWS deployment.

Week 5+: Explore self-hosting open-source models.